INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Third Semester Examination, November - 2024

Subject Code BCAAE301

Subject Name: Python Programming

Full Marks: 30 Duration: 1 Hour

Part - A

Attempt 5 questions

Each question carries 2 marks (2 × 5)

1. Explain the __init__() function 2

or

Write the Difference between __init__() and __str__() function 2

2. What is the functionality of lambda function in Python? Write the syntax of defining Lambda Function in Python. 2

or

Write a Python program to create a lambda function that adds 15 to a given number passed in as an argument, also create a lambda function that multiplies argument x with argument y and prints the result. 2

3. What is the output of the following list operation: 2

```
sampleList = [10, 20, 30, 40, 50]
print(sampleList[-2])
print(sampleList[-4:-1])
```

or

Write a program to create a tuple with different data types such as street address. 2

4. Why a list is said to be mutable? 2

or

Show how to initialize a list of tuples. 2

5. Write the differences between procedural programming and OOP's? Define Class, Object with an example using Python Syntax. 2

or

What are the four main principles of OOP? Explain the concept of Encapsulation. 2

Part - B

Attempt 2 questions

Each question carries 5 marks (5 × 2)

6. Write a function for checking whether the number is a prime number or not. 5

or

Write a Python program to sort a list of tuples using Lambda. **5**

Original list of tuples:
[('English', 89), ('Science', 90), ('Maths', 97), ('Social sciences', 80)]
Sorting the List of Tuples:
[('Social sciences', 80), ('English', 89), ('Science', 90), ('Maths', 97)]

7. **Explain the difference between *args and **kwargs with python code** **5**

or

Write a program to create a function show_employee() using the following conditions: **5**

1. It should accept the employee's name and salary and display both.
2. If the salary is missing in the function call then assign default value 9000 to salary

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Explain object oriented programming approach in detail? Write a Program to maintain salary of a worker using OOP. The details to receive are staff_id, basic_pay, years_of_experience, total_allowance and total_deduction with the help of Constructor. Using methods calculate net_pay. **10**

or

Write a Python program to create a class representing a shopping cart. Include methods for adding and removing items, and calculating the total price. Use the following instruction as work code flow for implementation: **10**

1) Define a class called ShoppingCart to represent a shopping cart.
2) Implement Add function to ADD an item with a name and quantity to the shopping cart
3) Implement REMOVE function to REMOVE an item with a specific name from the shopping cart.
4) Implement Calculate and return the total quantity of items in the shopping cart.
5) Justify your Program with Add items to the shopping cart
   - Items: ["Apple",120]; ["Guava", 60]; ["Grapes", 100]
   - Display the current items in the cart and calculate the total quantity
   - Remove an item from the cart, display the updated items, and recalculate the total quantity.